MILITARY SPECIFICATION

PROJECTILE, 60MM, SMOKE, WP, M302
METAL PARTS ASSEMBLY

This amendment forms a part of MIL-P-64036 (AR), dated 09 September 1986, and is approved for use by the U.S. Army Armament, Munitions and Chemical Command, and is available for use by all Departments and Agencies of the Department of Defense.

PAGE 2

2.1.1, Delete “MIL-STD-810...” in its entirety.

PAGE 3

2.2, Add “B117 Salt Spray (Fog) Testing” under the heading “American Society for Testing and Materials (ASTM)”.

PAGE 4

3.7.1, At end of paragraph add "See 4.4.5.5".

3.8, Delete the term "excess pressure" and substitute “maximum average pressure”.

PAGE 5

Add new paragraph as follows:

“3.11 Hydrotest. Each metal parts assembly shall successfully withstand an internal hydrostatic test load of 4300 + 200 psi. when tested in accordance with paragraph 4.4.3.6 without cracking, leakage, rupture or permanent deformation.”

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PAGE 7

TABLE I: In “Metal Parts Assembly (prior to paint)” section beneath “Examination for defects.. .” add:

“Hydrotest . ..10 . ..100% . ..3.11 . ..4.4.3.6”

In “No. of SAMPLE UNITS” column for the Salt Spray Test, delete “5” and substitute “3”.

Under “Salt Spray Test.. .” add:

“Paint Adhesion Test . ..2 . ..100% . ..3.2 . ..4.4.5.6.”

PAGE 8

4.4.2, Add the following to the end of the paragraph:

“Dimensions marked tool control (“*”) on Drawing 10534895 shall be gaged during First Article inspection, at the beginning of each workday, every workday every four (4) hours during production, for the last component prior to and first component after each tooling, equipment, or process change, repair, or adjustment. Measurements of the tool may be substituted provided the contractor has established correlation between the tool dimension and the component dimension prior to substitution. When acceptability has been determined, inspection of tool dimensions will be permitted. Whenever a defective tool control dimension is found, immediate action will be taken by the contractor to correct the tool problem and all material produced since the last successful inspection shall be rejected.”

PAGE 9

4.4.2.1, Delete in its entirety and substitute new 4.4.2.1. See CD sheet attached.

PAGE 11

4.4.2.2, Delete in its entirety and substitute new 4.4.2.2. See CD sheet attached.

PAGE 13

4.4.2.3, Delete in its entirety and substitute new 4.4.2.3.a - 4.4.2.3.d. See CD sheets attached.

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-P-64036 (AR)
Amendment 1

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 10534896
4.4.2.1	Adapter				NEXT HIGHER ASSEMBLY 10534894
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Special A	Metal defective		100%	3.6	Visual
Major					
101	Diameter of brazing surface		0.40%	3.2	Gage
102	Depth of brazing groove, max. (if present)		0.40%	3.2	Gage
103	Tensile test	2		3.2	4.4.3.1.4/Gage
104	Distance of brazing groove (if present) min., from shoulder and angle at rear		0.40%	3.2	Gage
105	Distance from rear face to shoulder		0.40%	3.2	Gage
106	Perpendicularity of shoulder to brazing surface		0.40%	3.2	Gage
107	Surface finish of brazing surface and shoulder		0.40%	3.2	Visual/6.1
Minor					
201	Rear outside diameter (OD)		0.65%	3.2	Gage
202	Length of angle at rear		0.65%	3.2	Gage
203	Angle rear of brazing surface		0.65%	3.2	Gage
204	Radii (except press-fit) or chamfers missing or incorrect		0.65%	3.2	Visual/6.1
205	Surface finish improper (in "250" areas)		0.65%	3.2	Visual/6.1
206	Evidence of poor workmanship		0.65%	3.10	Visual

NOTES:

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE	SHEET 1 OF 1	DRAWING NUMBER 10534895
4.4.2.2	Body		NEXT HIGHER ASSEMBLY 10534894

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Special A	Metal defective		100%	3.6	Visual
Major					
101	Diameter of brazing surface		0.40%	3.2	Gage
102	Length of brazing surface, min.		0.40%	3.2	Gage
103	Tensile test	1/		3.2	Gage/4.4.3.1.3
104	Angle at front OD		0.40%	3.2	Gage
105	Forward OD		0.40%	3.2	Gage
106	Surface finish of brazing surface and front face		0.40%	3.2	Visual/6.1
107	Radius at base of cavity	2/		3.2	Gage
108	Radius of blend in cavity, max.	2/		3.2	Gage
Minor					
201	Angle at rear of braze surface	2/		3.2	Gage
202	Radii or chamfers missing or incorrect		0.65%	3.2	Visual/6.1
203	Evidence of poor workmanship		0.65%	3.10	Visual

NOTES:

1/ Two (2) projectiles selected, four (4) specimens, total.
2/ See 4.4.2

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 1 OF 4		DRAWING NUMBER
4.4.2.3	Projectile, 60MM, Smoke, WP, M302				10534894 1/
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical					
1	Hydrotest		100%	3.11	4.4.3.6/Gage
2	Air leak test		100%	3.2	4.4.3.2/Gage
3	Diameter of press-fit surface improper		100%	3.2	Gage
4	Improper taper on press-fit surface		100%	3.2	Gage
5	Improper surface finish of press-fit surface or its radius		100%	3.2	Gage
Special					
A	Width of gas check grooves (4 places)		0.40%	3.2	Gage
B	Depth of gas check grooves (4 places)		0.40%	3.2	Gage
C	Braze metal not visible around entire periphery of joint		100%	3.7	Visual
D	Minor diameter of rear thread, max.		100%	3.2	Gage
E	Braze joint defective	5		3.7.1	4.4.3.3/4.4.5.5 Visual
F	Improper blend of radius to press-fit surface		100%	3.2	Visual
G	Radius at start of press-fit area missing or improper anywhere on circumference		100%	3.2	Visual/6.1

NOTES: 1/ Refer to component drawing when characteristic is not shown on Dwg. 10534894.

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.3	Projectile, 60MM, Smoke, WP, M302	2 OF 4	10534894 1/
			NEXT HIGHER ASSEMBLY

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Major					
101	True position of fuze (forward) thread pitch diameter (PD) with bourrelet and front face (2 locations)		0.40%	3.2	Gage
102	Runout of body OD with bourrelet		0.40%	3.2	Gage
103	Runout of press-fit surface with bourrelet		0.40%	3.2	Gage
104	Runout of adapter taper with bourrelet		0.40%	3.2	Gage
105	Runout of rear taper with bourrelet		0.40%	3.2	Gage
106	True position of rear thread PD with bourrelet and rear face (2 locations)		0.40%	3.2	Gage
107	Presence of flux, oxides, or foreign matter in cavity or on exterior		0.40%	3.2	Visual
108	Protective finish missing or inadequate		0.40%	3.2	Visual
109	Depth from front face to bottom of fuze cavity		0.40%	3.2	Gage
110	Fuze thread PD		0.40%	3.2	Gage
111	Fuze thread minor diameter		0.40%	3.2	Gage 2/
112	OD rear of bourrelet		0.40%	3.2	Gage
113	Bourrelet diameter		0.40%	3.2	Gage
114	Length to outside basic diameter from rear face		0.40%	3.2	Gage

NOTES:
1/ Refer to component drawing when characteristic is not shown on Dwg. 10534894.
2/ Minor diameter shall be checked to full form min. length.

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 3 OF 4		DRAWING NUMBER 10534894 1/
4.4.2.3	Projectile, 60MM, Smoke, WP, M302				NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO / OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Major					
115	Large outside diameter of fuze adapter		0.40%	3.2	Gage
116	Length from rear face to base of cavity		0.40%	3.2	Gage
117	Wall thickness in tapered zone		0.40%	3.2	Gage
118	Sharp corners not present at front or rear edges of gas-check grooves or at rear of bourrelet		0.40%	3.2	Visual
119	Rear thread PD		0.40%	3.2	Gage
120	Distance of gas-check grooves from rear groove		0.40%	3.2	Gage
121	Surface finish of gas-check OD		0.40%	3.2	Visual/6.1
122	Wall thickness rear of gas-check grooves		0.40%	3.2	Gage
123	Radius of blend at front of taper on OD		2/	3.2	Gage
124	Distance from front face to rear of bourrelet		0.40%	3.2	Gage
125	Distance from front face to rear gas-check groove		0.40%	3.2	Gage

NOTES: 1/ Refer to component drawing when characteristic is not shown on Dwg. 10534894.
2/ See 4.4.2

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 4 OF 4		DRAWING NUMBER
4.4.2.3	Projectile, 60MM, Smoke, WP, M302				10534894 1/
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Minor					
201	Marking missing, incorrect or illegible		0.65%	3.2	Visual
202	Forward OD		0.65%	3.2	Gage
203	Diameter of chamfer at front face		0.65%	3.2	Gage
204	Rear OD		0.65%	3.2	Gage
205	Diameter at rear chamfer		0.65%	3.2	Gage
206	Diameter of rear undercut		0.65%	3.2	Gage
207	Width of rear undercut		0.65%	3.2	Gage
208	Depth of rear thread cavity		0.65%	3.2	Gage
209	Angle at rear face		0.65%	3.2	Gage
210	Minor diameter of rear thread, min.		0.65%	3.2	Gage
211	Radius at rear of bourrelet		2/	3.2	Gage
212	Air pressure or hydrostatic test stamps missing		100%	3.2	Visual
213	Radius, max., at bottom of gas-check grooves (4 places)		0.65%	3.2	Gage
214	Location of marks (3 places)		0.65%	3.2	Gage
215	Evidence of poor workmanship		0.65%	3.2	Visual

NOTES:
1/ Refer to component drawing when characteristic is not shown on Dwg. 10534894.
2/ See 4.4.2

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PAGE 15

4.4.2.4, Major 107: Delete “max.”

Major 108: Delete “5” and substitute “3” in “NO. OF SAMPLE UNITS” column.

Add Major “109 . . .Paint adhesion test . . .2 . . .- . . . .3.2 . . .4.4.5.6/Visual.”

Minor 201: Delete callout in its entirety and substitute, “Paint or wash-primer present on front or rear surfaces or inside cavity.”

PAGE 18

4.4.3.1, Delete in its entirety and substitute the following:

“4.4.3.1 Adapter. Each month one (1) tension test specimen shall be taken from each of two (2) bars representative of each mill heat used during that month’s production to determine compliance with the yield strength and elongation requirements of the applicable drawing. If one (1) mill heat is sufficient for more than one month’s production the test shall be repeated each month, or the total number of specimens required during that period of production may be tested at the beginning of that mill heat’s use. Failure of any specimen to comply with either the yield strength or elongation requirements shall cause rejection of the applicable mill heat. See 4.4.5.3.2.”

4.4.3.3, Delete in its entirety and substitute the following:

"4.4.3.3 Joint quality. The test shall be performed for First Article inspection, each day of brazing during production and whenever a change is made to the brazing process. Three (3) samples shall be tested at the start of brazing operations and after a layoff of over one (1) hour, and two (2) shall be tested at the end of brazing operations during continuous brazing up to 24 hours, or when a layoff of over one (1) hour will occur. All samples shall be sectioned and peeled in accordance with 4.4.5.5 to determine compliance with 3.7.1. All specimens must pass. If a failure occurs in either of the two (2) samples tested at the end of the day’s brazing operations or at the start of layoff, all parts brazed since the last successful test shall be rejected.”

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4.4.3.5, Delete the first sentence in its entirety and substitute "The samples supplied in accordance with Table I and 4.4.2.4 shall be tested as specified in 4.4.5.7."

Table II, Delete "1 2" in Accept/Reject column and substitute "0 1"

Add: "4.4.3.6 Hydrotest. After the wall thickness is in conformance with final size but prior to the air leak test, each projectile assembly shall be subjected to an internal hydrotest to determine compliance with the requirements of 3.11. Immediately after removal from the hydrotest equipment, each projectile that has successfully passed the hydrotest shall be stamped "H" 1/8 inch high by .01 inch deep approximately 1/8 inch below the location of the zone used for the air leak stamp. Projectile bodies shall be tested using equipment capable of performing the test accurately and consistently. During the test the pressure will be applied gradually until the required minimum pressure is reached and held for five (5) seconds, minimum. The equipment shall be calibrated prior to the start of each day's operation and at intervals of not over 4 hours of continuous operation. It shall also be calibrated prior to restarting the operation after a lay-off of three hours or more. If it has been found that the equipment is out of calibration in such a direction that the pressure was actually lower than required by the test, the bodies tested since the last satisfactory calibration shall be retested after the calibration error has been corrected. Projectile rupture, cracking, leakage or bulging during the test shall be cause for rejection of the body. The test fixture shall not restrict the projectile body from expanding longitudinally or radially under pressure. Protective barriers should be installed where applicable to shield operators and material from metal fragments which may come from defective projectiles that fail the test. If threads are engaged, the hydrotest plug should be designed with a one quarter inch minimum thread relief, to prevent loading of initial threads on the projectile body. Examination should be made of all hydrotest plugs after a projectile is blown during testing. Badly nicked or damaged plugs may cause destruction of acceptable items."

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4.4.5.5, Paragraph c, Add: "Examination shall be accomplished by peeling back and removing the body segments from the adapter segments. In addition to the joint quality requirements of 3.7.1, the braze metal on all surfaces must exhibit a uniform sandy appearance wherever it is present. Smooth braze metal or other evidence of non-adhesion on any segment shall cause rejection of all material brazed since the last successful inspection."

4.4.5.6, Delete in its entirety and substitute the following:

"4.4.5.6 Salt spray test. See 4.4.3.4. In performing the following tests, TT-C-490 shall apply unless specified otherwise herein. The test shall be performed for first article inspection, each day of painting during production and whenever a change is made to the paint process, until all samples from five consecutive days of painting have passed the test. Thereafter the sample frequency may be reduced to twice per week during continuous days of painting. Five specimens shall be selected, consisting of randomly chosen projectiles which have been processed through all the same cleaning, phosphating, painting and drying steps at the same time using the same equipment as other production projectiles. The samples shall be taken from each continuous final paint operation up to 24 hours long. One projectile shall be taken during the first thirty (30) minutes of painting, one during the last thirty (30) minutes and the others at any other time.

Prior to salt spray testing, two (2) of the specimens must only pass an adhesion test using the procedures and pass-fail criteria contained in 4.2.8 and 3.5.6 respectively, of TT-C-490. If both specimens pass the adhesion test, the three (3) remaining specimens shall be scored lengthwise using a precision scribing tool/insert or equivalent in preparation for the salt spray test.

The salt solution, test equipment and drying procedures shall be in accordance with ASTM B117, Salt Spray (Fog) Testing, unless specified otherwise herein. Specimens shall be positioned between 15 and 30 degrees from vertical during the test. No coating of any kind shall be permitted over the paint on stamp marks or gas-check grooves of the specimens prior to testing, however areas in contact with racks or supports, unpainted surfaces and areas within 3/16 inch of unpainted surfaces may be protected.

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When examining the specimens for signs of paint failure, the entire surface of the projectile shall be considered the significant surface.

As used herein, the term "blister" refers to an unbroken area of paint with a bubble-like appearance whereas "pits" or "spots" are exposed phosphate pretreatment or bare metal visible through holes in the paint. Blisters which are present when the test is over are to be counted within five (5) minutes of removal from the test chamber.

Projectiles shall be tested for the duration specified on Drawing 10534894. After testing, any projectile which fails for any of the following reasons shall cause rejection of all parts painted since the last successful test and a return to the original sampling plan if failure occurred during reduced sampling. The failure criteria in a-i applies to the significant surface only.

a. The paint shows more than 1/8 inch creepage, blistering or loss of adhesion of paint from the scribe mark, stamp mark, or edge.

b. There are more than four (4) blisters, total.

c. Any blister exceeds 1/16 inch average diameter.

d. Any pit exceeds 1/8 inch average diameter.

e. There are three (3) or more scattered pits exceeding 1/16 inch average diameter.

f. There are three (3) or more spots within one square inch.

g. There are two (2) spots within one square inch in two (2) or more places.

h. There are more than four (4) scattered spots, total.

i. There is evidence of rust under the paint (a brown discoloration showing through the paint) in excess of 1/4 inch in diameter total area.

j. There is some variation of the above or an unlisted defect anywhere which the Government representative deems excessive."

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PAGE 21

4.4.5.7.1, Delete “Excess pressure” and substitute “Maximum average pressure.”

6.3, Delete “Dover, NJ 07801-5000” and substitute “Picatinny Arsenal, NJ 07806-5000.”

6.4, Delete “Dover, NJ 07801-5000” and substitute “Picatinny Arsenal, NJ 07806-5000.”

Custodian:
Army-AR

Preparing activity:
Army-AR

(Project 1310-A539)

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